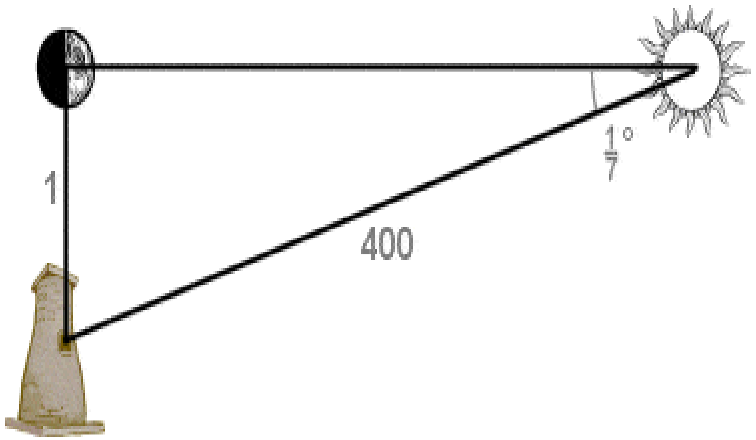
Chapter 8: From Brahmagupta to Bhaskara II to Narayana

Mathematics Moves South

- **Aryabhata's Influence** Spread Rapidly Within a generation, his ideas reached Varahamihira in Ujjain (1000+ km away). His work introduced trigonometric functions particularly the sine table which appeared across the *Pancasiddhantika*. His geometry treated Earth as a rotating sphere, revolutionary for its time.
- **Brahmagupta's Diophantine Equation** He pioneered solving equations of the form $x^2 - Ny^2 = 1$ in integers. Bhaskara II later solved it fully using the Chakravala (cyclic) method.
- **Kerala's Precise Sine Tables** Shankaranarayana used actual sine values (not differences), making astronomical calculations far more accurate. This tradition later led to Madhava's infinite series.
- **Mathematics Moves South** Ghazni's invasions (1000 CE) scattered scholars southward. Kerala became the new hub, where Madhava later developed infinite series for π, predating European calculus by 200 years.



Indian trigonometry tables showed that an angle of 1/7° indicates triangle sides with a ratio of 400:1, meaning that the Sun is 400 times further away from the Earth than the Moon

- **Bhaskara II The Great Unifier** Born 1114 CE, his works *Lilavati* and *Bijaganita* unified arithmetic, algebra and astronomy into one coherent system, marking a major point of completion in Indian mathematics.

The Quadratic Diophantine Problem - bhāvanā

- **Brahmagupta's Invention:** Brahmagupta invented the equation $Nx^2 + C = y^2$ where N is a non-square positive integer and C is any integer. The goal was to find integer solutions for x and y. No Indian mathematician before him attempted this.
- This is called vargaprakṛti ("of square nature"). The goal is to find positive integers x (first root) and y (second root) satisfying the equation for a given multiplier N and interpolator C.

Example: Take $N = 2$, $C = 1$ → solve $\mathbf{2x^2 + 1 = y^2}$

- Try $x = 2$: $2(4) + 1 = 9 = 3^2$ → Solution: $(x, y) = (2, 3)$
- Try $x = 12$: $2(144) + 1 = 289 = 17^2$ → Solution: $(x, y) = (12, 17)$
- **Bhāvanā - The Composition Law:** Brahmagupta's key discovery (bhāvanā) states: if (x_1, y_1) solves $Nx^2 + C_1 = y^2$ and (x_2, y_2) solves $Nx^2 + C_2 = y^2$, then composing them gives a new solution.
- **The Rule:** $(x_1, y_1) \times_+ (x_2, y_2) = \mathbf{(x_1y_2 + x_2y_1, y_1y_2 + Nx_1x_2)}$ This new pair solves $\mathbf{Nx^2 + C_1C_2 = y^2}$
- **Example:** For $N = 2$: $(2, 3)$ solves $2x^2 + 1 = y^2$ Compose with itself: $x = 2\cdot3 + 3\cdot2 = \mathbf{12}$, $y = 3^2 + 2\cdot(2^2) = 9 + 8 = \mathbf{17}$ → Check: $2(144) + 1 = 289 = 17^2$ ✓ (New, larger solution generated!)

The Quadratic Diophantine Problem - bhāvanā

- **The Special Equation - Pell's Equation:** The most important case is $C = 1$, known today as Pell's equation: $\mathbf{Nx^2 + 1 = y^2}$. Brahmagupta showed that if one solution exists, infinitely many can be generated via bhāvanā.
- If (x, y) is a solution, composing it with itself gives $(2xy, y^2 + Nx^2)$, also a solution.
- **Example:** $N = 3$: solve $\mathbf{3x^2 + 1 = y^2}$
- $x = 1$: $3(1) + 1 = 4 = 2^2 \quad \rightarrow (x, y) = (1, 2)$
- Compose $(1, 2)$ with itself: $x' = 2(1)(2) = \mathbf{4}$, $y' = 4 + 3(1) = \mathbf{7}$
- Check: $3(16) + 1 = 49 = 7^2 \quad \rightarrow (4, 7)$ is the next solution!
- Next: compose $(4, 7) \rightarrow x'' = 56$, $y'' = 97 \rightarrow 3(3136) + 1 = 9409 = 97^2$
- **The Cakravāla Algorithm - The Cyclic Method:** Jayadeva (11th century), later described by Bhaskara II, invented *cakravāla* ("cyclic algorithm") — a systematic method to solve Pell's equation by linking it to the linear equation (kuṭṭaka) repeatedly until $C = \pm1, \pm2$, or ±4 is reached.
- **The Core Idea:** Choose an integer m such that $(m^2 - N)/C$ gives a small new interpolator. Keep cycling until $C = 1$.

The Quadratic Diophantine Problem - bhāvanā

- **Steps of Cakravāla**:

1. Start with a trial solution $(x_1, y_1; C_1)$
2. Find integer m minimising $|m^2 - N|$ such that $C_1 \mid (m + y_1)$
3. Compose to get new triple → repeat until $C = 1$

Example: $N = 67$, start: $x=1$, $y=8$, $C = 64-67 = -3$ After cycling through several steps (Bhaskara's own example), the minimal solution emerges: → **(x, y) = (5967, 48842)**, i.e., $67 \times 5967^2 + 1 = 48842^2$

- **Legacy - From Brahmagupta to Modern Number Theory:** Brahmagupta's bhāvanā is a **binary composition law on the space of solutions**, studied abstractly centuries before Europe. Fermat (1657 AD) posed Pell's equation as a challenge; Euler and Lagrange solved it ~100 years later using continued fractions the same structural ideas Brahmagupta had in 628 AD.

Methods of Solution – cakravāla
Rational Solutions and Scaling Method

- **Brahmagupta realized**:Fractional (rational) solutions can help us find integer solutions.
- Auxiliary Rule: If $C = k^2M$ then: $Nx^2 + C = y^2$ can be reduced to: $Nx^2 + M = y^2$ by dividing through by k^2.
- **Example:** Equation: $3x^2 - 800 = y^2$ Since:$800 = 20^2 \times 2$ divide by 400 : $3\left(\frac{x}{20}\right)^2 - 2 = \left(\frac{y}{20}\right)^2$
- Now solve the simpler equation:$3x^2 - 2 = y^2$, x=1,y=1 Check:$3(1)^2 - 2 = 1$
- Multiply back by 20:$x = 20, y = 20$, Check original equation:$3(20)^2 - 800 = 1200 - 800 = 400 = 20^2$

This method shows:

1. Large difficult equations can be transformed into smaller easier equations.
2. Scaling is an important algebraic tool even today.

Converting Fractional Solutions into IntegerType equation here. Solutions

- **Bhāvanā can transform**:**fractional solutions → integer solutions**, This is one of the most beautiful parts of Brahmagupta's mathematics.
- **Example:**Equation: $92x^2 + 1 = y^2$ Start with: $92(1)^2 + 8 = 10^2$ So:$(1, 10)$

solves:$92x^2 + 8 = y^2$

- **Step 1 – Compose the Solution** Using Bhāvanā: $(1, 10)_+^2 = (20, 192)$ This solves:$92x^2 + 64 = y^2$
- Now divide by 64:$\left(\frac{5}{2}, 24\right)$,This becomes a fractional solution of:$92x^2 + 1 = y^2$
- **Step 2 – Compose Again**:Apply:$(x, y)_+^2 = (2xy, 2y^2 - 1)$ Substitute:$x = \frac{5}{2}, y = 24$
- Then:$x = 2 \cdot \frac{5}{2} \cdot 24 = 120$ $, y = 2(24)^2 - 1$ $= 1152 - 1 = 1151$
- Final integer solution:$(x, y) = (120, 1151)$
- Brahmagupta did not reject fractions.
- He used them strategically.
- This is similar to modern mathematical transformations.

Special Cases $C = -1, \pm 2, \pm 4$

- Certain special values of C help generate solutions faster.These are:$C = -1, \pm 2, \pm 4$
- Indian mathematicians discovered special formulas for them.
- Example: Case $C = -1$,Suppose:$Nx^2 - 1 = y^2$has a solution. Then Bhāvanā gives a solution for:$Nx^2 + 1 = y^2$
- Example:Take: $2(1)^2 - 1 = 1$, So(x,y)= (1,1)
- solves: $2x^2 - 1 = y^2$ Now compose:$(x, y)_+^2 = (2xy, Nx^2 + y^2)$ Substitute values: $x' = 2(1)(1) = 2$, $y' = 2(1)^2 + 1^2 = 3$, So:$(x, y) = (2, 3)$

This shows:

1. Ancient Indian mathematicians classified equations into special categories.
2. Different algebraic tricks were used for different cases.

Cakravāla Method (Cyclic Algorithm)

- Cakravāla is a systematic algorithm developed mainly by: 1. Jayadeva 2. Bhaskara II , It solved equations like: $Nx^2 + 1 = y^2$ efficiently.
- Main Principle: Start with: $Nx_0^2 + C_0 = y_0^2$, Then repeatedly reduce the value of C , Goal: $C = 1$
- Important Step: Choose v such that:: $x_1 = \frac{vx_0 + y_0}{C_0}$ becomes an integer.
- Then compute: $y_1 = \frac{Nx_0 + vy_0}{C_0}$ and $C_1 = \frac{v^2 - N}{C_0}$ This creates a new simpler equation.
- Cakravāla was one of the greatest achievements of medieval mathematics because it combined:

1. recursion
2. algebra
3. number theory
4. systematic computation

long before similar European methods appeared.

A Different Circle Geometry: Cyclic Quadrilaterals

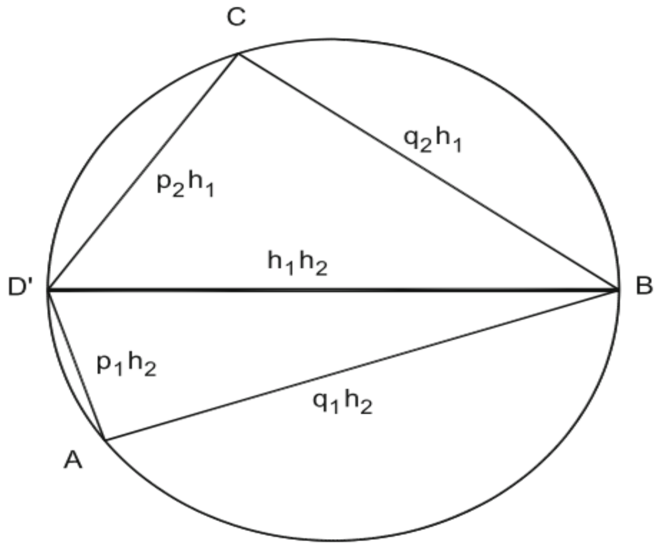
1. **A cyclic quadrilateral:** is a quadrilateral whose four vertices lie on the same circle.**Important Property**

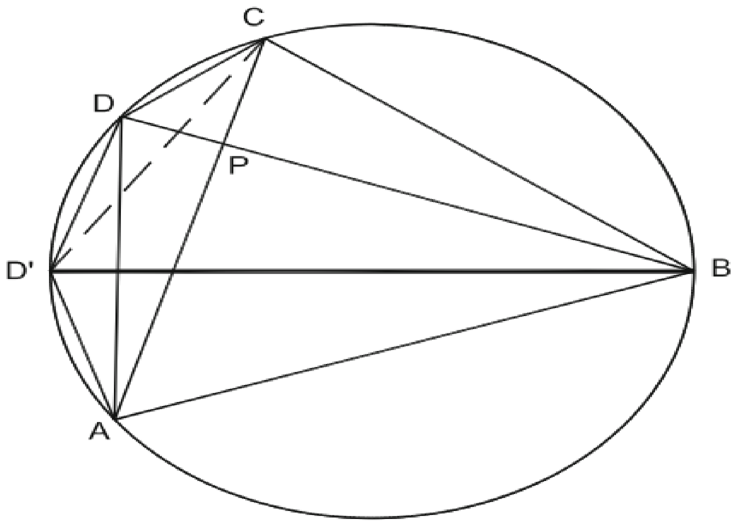
- Opposite angles are supplementary:$\angle A + \angle C = 180°$.
- This property was the foundation of Brahmagupta's study of quadrilaterals.

2. **Construction of Brahmagupta Quadrilateral:**Brahmagupta combined two right triangles having the same hypotenuse to form a cyclic quadrilateral. If the right triangles are:$(h_1; p_1; q_1), (h_2; p_2; q_2$ then the sides of the quadrilateral become:$p_1h_2,\ q_1h_2,\ p_2h_1,\ q_2h_1$ The diagonals intersect at right angles: $d_1 \perp d_2$. This makes the figure both geometrically elegant and numerically useful.

- **Brahmagupta's Area Formula:** For a cyclic quadrilateral with sides:a_1, a_2, a_3, a_4 and semiperimeter:$s = \frac{a_1+a_2+a_3+a_4}{2}$

the area is:$A^2 = (s - a_1)(s - a_2)(s - a_3)(s - a_4)$





A Different Circle Geometry: Cyclic Quadrilaterals

4. **Diagonal Theorem:** Brahmagupta gave formulas for the diagonals of a cyclic quadrilateral. For diagonals d_1 and d_2: $d_1 d_2 = a_1 a_3 + a_2 a_4$

- This is closely related to **Ptolemy's theorem**.
- **In Brahmagupta Quadrilateral:** Area can also be written as: $A = \frac{1}{2} d_1 d_2$ because the diagonals are perpendicular.

5. Historical Significance

- Brahmagupta connected **geometry with algebra and number theory**.
- His work influenced later Indian mathematicians such as:

1. Bhaskara II
2. Narayana Pandita

- His methods later contributed to advanced studies of cyclic quadrilaterals and trigonometry in Indian mathematics.

The Third Diagonal; Proofs

1. Narayana Pandita extended Brahmagupta's ideas on cyclic quadrilaterals. He introduced a new concept called the **third diagonal**.

- For a cyclic quadrilateral: First diagonal: $d_1 = AC$, Second diagonal: $d_2 = BD$, Third diagonal: $d_3 = BD'$.
- The third diagonal appears after transposing (interchanging) sides of the quadrilateral.

2. **Transposition of Sides**:Narayana studied how cyclic quadrilaterals change when two sides are interchanged.
Key Observation: The **area remains unchanged** under transposition.
If $A$is the area: $A(ABCD) = A(ABCD')$
This idea became central in proving many results about cyclic quadrilaterals.

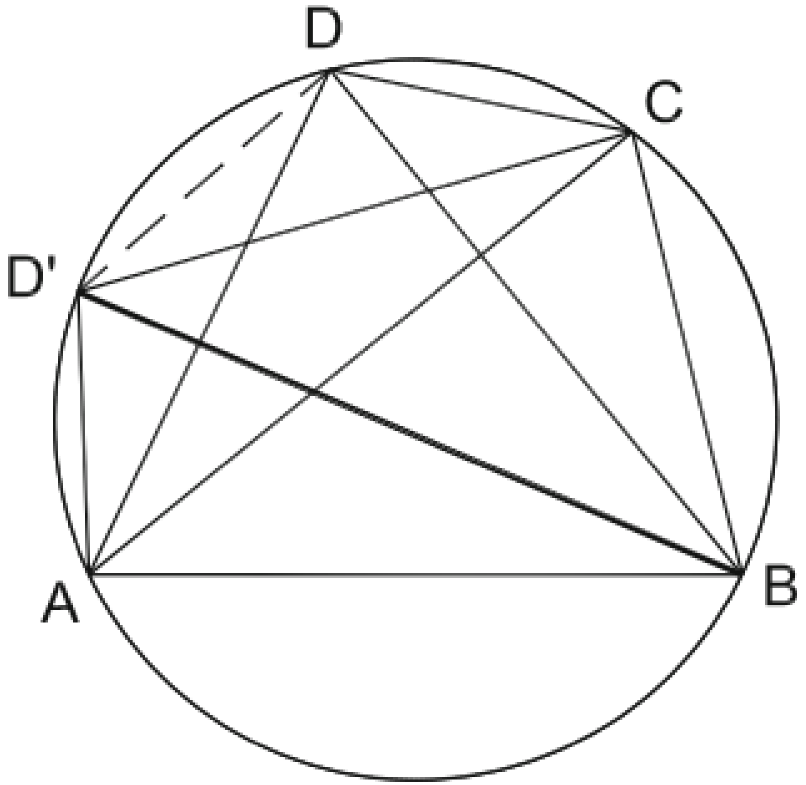


Fig : Transposition and third diagonal

The Third Diagonal; Proofs

3. Ratio of Diagonals:

- Narayana derived elegant relationships between side For diagonals $d_1$and d_2:

$$\frac{d_1}{d_2} = \frac{a_1a_4 + a_2a_3}{a_1a_2 + a_3a_4}$$

- The ratio of diagonals depends only on the sides of the cyclic quadrilateral.This generalized earlier ideas of Brahmagupta. Type equation here.

4. Narayana's Beautiful Area Theorem: Narayana proved that the area of a cyclic quadrilateral can also be written using the three diagonals and the circumdiameter D: $2AD = d_1d_2d_3$

where: A= area, D = diameter of circumcircle, Type equation here.d_1,d_2,d_3 = three diagonals

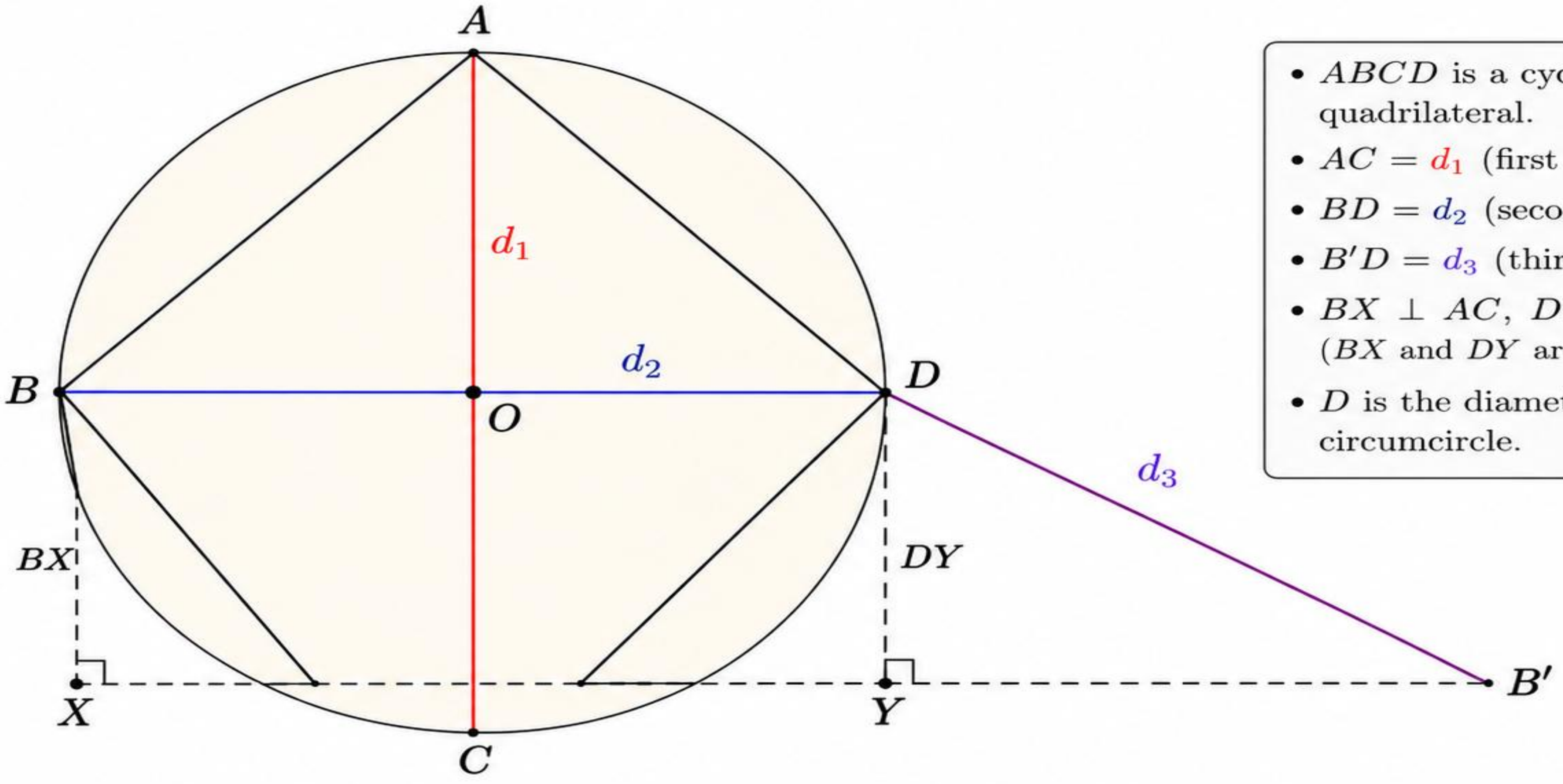
- **Special Case:**If one diagonal becomes the diameter, the formula reduces to the Brahmagupta quadrilateral case.

5. Contribution and Legacy: Narayana completed and generalized Brahmagupta's geometric program.

- His methods later helped mathematicians prove:1. Brahmagupta's Area Formula,2. Diagonal Theorem
- His work influenced later Kerala mathematicians and the development of advanced Indian geometry and trigonometry.

Figure: Proof of Narayana's Theorem

$$2AD = d_1 d_2\, d_3$$



- $ABCD$ is a cyclic quadrilateral.
- $AC = d_1$ (first diagonal)
- $BD = d_2$ (second diagonal)
- $B'D = d_3$ (third diagonal)
- $BX \perp AC$, $DY \perp AC$ (BX and DY are altitudes)
- D is the diameter of the circumcircle.

Then $BX + DY$ is an altitude of triangle $DB'B$ to the side DB' which has the same circumdiameter D as the given circle. Hence,

$$BX + DY = \frac{BD \cdot DB'}{D} = \frac{d_2 d_3}{D}.$$

$\Rightarrow$

Area of $ABCD$

$$A = \frac{1}{2} AC\ (BX + DY) = \frac{1}{2} d_1 \cdot \frac{d_2 d_3}{D} = \frac{d_1 d_2 d_3}{2D}$$

$\Rightarrow$

$$2AD = d_1 d_2 d_3$$

(Narayana's Theorem)

Proof of Narayan's Theorem

$2AD = d_1 d_2 d_3$

A = Area of the cyclic quadrilateral

D = Diameter of the circumcircle

d_1, d_2, d_3 = the three diagonal introduces by Narayan

1. Consider a cyclic quadrilateral ABCD

its diagonals are

$AC = d_1$

$BD = d_2$

After transposing one side Narayan introduced a third diagonal.

$B'D = d_3$

Draw Perpendicular BX and DY as altitudes.

(2) Area of the quadrilateral

The quadrilateral is divided two triangles by diagonal AC.

So, $A = \frac{1}{2} AC (BX + DY)$

since $AC = d_1$

$$A = d_1 \left(\frac{BX + DY}{2} \right)$$

$$\underline{BX + DY = \frac{d_2 d_3}{D}}$$

↓

This comes from properties of similar triangles and the circumcircle.

$$A = \frac{d_1 (BX + DY)}{2}$$

$$A = \frac{d_1}{2} \cdot \frac{d_2 d_3}{D}$$

$$A = \frac{d_1 d_2 d_3}{2D}$$

$$\boxed{2AD = d_1 d_2 d_3}$$